

Note - Homework 1 - Arduino Photoresistor Maze

Problem:

The goal of this task was to build a small non-trivial prototype using the virtual Arduino environment Tinkercad.

I made a small interactive maze/puzzle game, where the player, enemies and the game map are displayed in a 6x6 LED matrix and represented by individual colored pixels. The design also features a photoresistor that controls which map is shown depending on light level. The player must move between the two maps to reach the green LED.

Design:

Prototype features input in form of 5 buttons connected to ports 9-13 that control the player and display the player's health, as well as a photoresistor connected to the A0 port that lets you change between light and dark maps.

Output consists of a LED matrix made out of 6 LED strips stacked into a 6x6 pixel grid, connected linearly from the bottom row to the top row and connected to port 5. Also, a 16x2 LCD screen to show simple instructions, health, report damage, loss or winning and a piezo connected to port 3 for simple sounds and melodies.

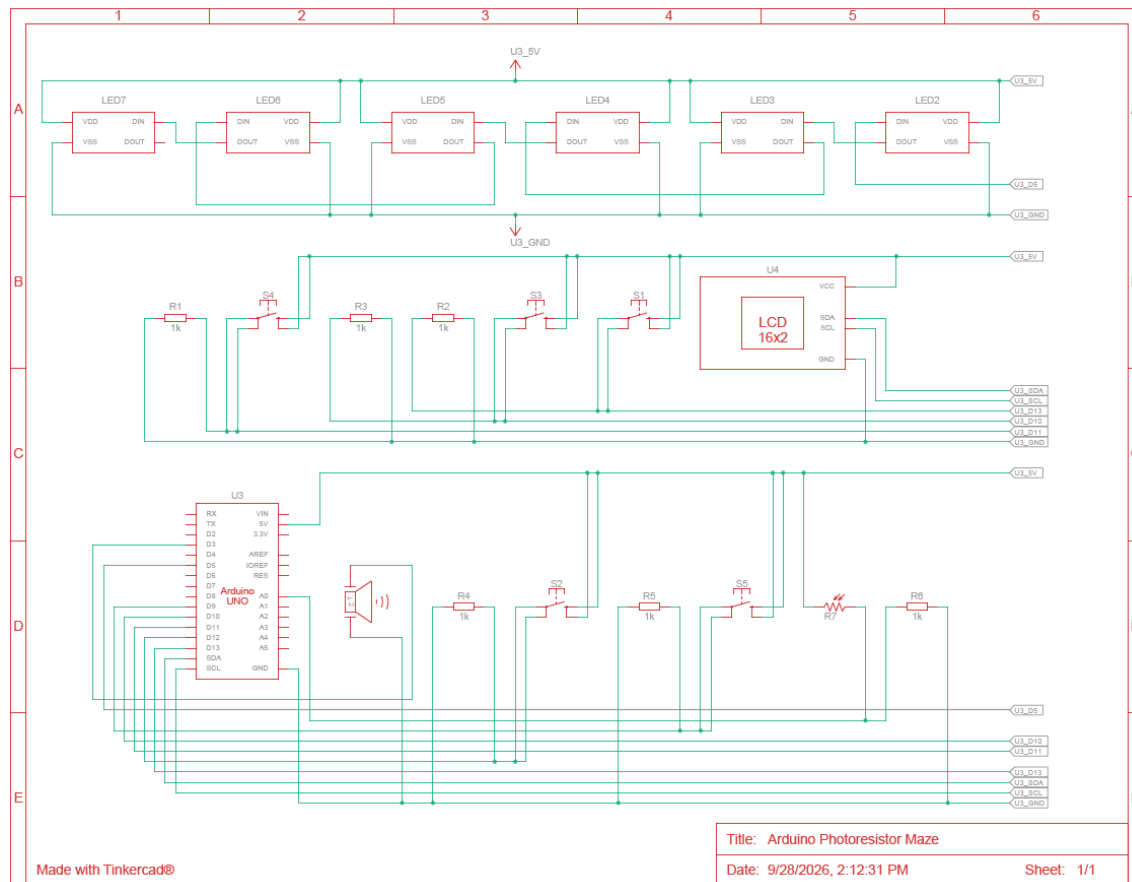
The program consists of several different classes for controlling inputs and outputs (ButtonController, SoundController, StripController, LcdController). Then there are classes CollisionMap, EnemyMap, Player that manage the game logic such as what and when needs to be displayed (in the LED matrix and LCD screen), which movements are valid and what are their consequences, player health.

Parts:

- 1 Arduino Uno R3
- 1 MCP23008-based, 32 (0x20) LCD 16 x 2 (I2C)
- 6 NeoPixel Strip 6
- 1 Breadboard Small
- 5 Pushbutton
- 6 1 k Ω Resistor
- 1 Photoresistor

1 Piezo

Schematic:



What works and doesn't:

The main parts of the prototype work well, while still lacking some polish.

Input and output processing, player movements, collision detection, health system with enemy damage, map changes, winning, losing, various sounds and short melodies all work as intended.

The photoresistor functionality works but could easily be replaced with more practical input options. This game feature lacks substance and reasonable impact on the gameplay. It is also unknown how valid this kind of control would be in a real environment as the light threshold is fixed at 512.

The Tinkercad environment is slow and unstable, making development and testing difficult.

Future improvements:

The game could feature more complex gameplay and better use of the photoresistor's capabilities.